The algorithm operates in three stages. First, geomagnetic storms are detected over the time period that the user has set. Second, the detected events are filtered, and their main phases are identified. Third, derived solar wind parameters are extracted for the filtered storms with data-complete main phases.

Stage 1: Storm Detection

The first stage identifies geomagnetic storms based on their Dst minima. At this stage, the program is concerned only with the minimum Dst depression of each storm and the time at which this minimum occurs.

<u>Stage 1 Algorithm parameters:</u>

EPISODE_LEVEL defines the Dst threshold used to identify continuous disturbed intervals. A disturbed episode begins when Dst becomes equal to or lower than EPISODE_LEVEL and continues as long as consecutive hourly Dst values remain at or below this threshold. Thus, EPISODE_LEVEL acts as the initial, broad criterion for identifying magnetically disturbed periods. The default value is **−20 nT**.

STORM_LEVEL defines the Dst threshold used to identify storm-level minima within each disturbed episode. After a disturbed episode has been detected, the algorithm searches within that episode for candidate storm minima. A **candidate storm minimum** is defined as a Dst value that is equal to or lower than STORM_LEVEL and is also a local minimum within the surrounding time window defined by LOCAL_MIN_RADIUS_HOURS. The default value is **−50 nT**, which is commonly used as the threshold for identifying moderate geomagnetic storms.

LOCAL_MIN_RADIUS_HOURS defines the time window used to test whether a storm-level Dst value is a local minimum. For each Dst value that is equal to or lower than STORM_LEVEL, the algorithm checks whether it is the lowest Dst value within the surrounding ±LOCAL_MIN_RADIUS_HOURS window. If it is, the point is retained as a candidate storm minimum. The default value is **3 hours**.

MINIMA_SPLIT_FACTOR defines the **minimum-splitting criterion**, which controls whether multiple candidate storm minima within the same disturbed episode are treated as separate storm events. When two candidate storm minima occur within the same episode, the algorithm examines the maximum Dst value between them, referred to here as the inter-minimum peak. The inter-minimum peak height is defined as:

peak height = inter-minimum peak − shallower minimum

This height is then compared with a separation threshold, defined as a fraction of the absolute depth of the deepest minimum:

separation threshold = −deeper minimum × MINIMA_SPLIT_FACTOR

If the peak height is greater than or equal to the separation threshold, the two minima are treated as separate storm events. Otherwise, they are considered part of the same storm structure, and the deeper minimum is retained as the representative minimum Dst of the storm. Default value for MINIMA_SPLIT_FACTOR is **0.4**.

If more than two candidate storm minima occur within the same episode, they are evaluated sequentially in chronological order. Each new candidate minimum is compared with the most recently retained minimum. If the separation criterion is satisfied, the new minimum is retained as a separate storm event. If not, the two minima are merged and only the deeper minimum is retained before the algorithm proceeds to the next candidate minimum.

STORM_LIMIT imposes a lower bound on accepted Dst minima. It allows the user to restrict the analysis to a selected storm-intensity range and to exclude events outside the desired limits.

Note: To select a specific storm-intensity range, STORM_LEVEL can be used as the upper threshold and STORM_LIMIT as the lower threshold. For example, setting STORM_LEVEL = −50 nT and STORM_LIMIT = −99 nT restricts the analysis to storms with −99 nT ≤ Dst_min ≤ −50 nT (moderate storms).

Stage 2: Main-Phase Identification

In the second stage, the program identifies the main phase of each storm detected in Stage 1. Because the end of the main phase is defined by the minimum Dst time, t_min, Stage 2 focuses on estimating the corresponding onset time of the main phase, t_start.

For each detected storm, the algorithm searches backward in time from t_min. This search uses a reference Dst value, referred to as the **mark**. Once the backward search reaches a Dst value equal to or higher than the mark, the first local Dst maximum encountered is assigned as Dst_start, and its timestamp is assigned as t_start. The main-phase duration is then calculated as the time difference between t_start and t_min.

The Stage 1 storm detection parameters are not reapplied during this stage. Stage 2 uses only the storm minima and their timestamps previously identified during storm detection. Consequently, the determination of t_start is not constrained by the temporal limits of the original disturbed episode. Instead, the algorithm searches backward through the available Dst record until both the mark condition and the local maximum criterion are satisfied. If no valid t_start can be identified, the storm is excluded from the filtered storm sample. In this study, filtered storms are defined as detected storms for which the main phase can be reliably identified.

<u>Stage 2 algorithm parameters:</u>

STRONG_STORM_THRESHOLD separates strong storms from weaker storms for the purpose of defining the mark that will be used for main phase onset identification. The default value is **−100 nT**. Storms with Dst_min equal to or deeper than this threshold use the fixed mark value defined by STRONG_MARK.

STRONG_MARK is the mark used for strong storms. The default value is **−50 nT**. During the backward search from t_min, once Dst reaches or exceeds this value, the first local maximum encountered is assigned as Dst_start, and its timestamp is assigned as t_start.

For weaker storms, the mark is scaled according to storm intensity:

WEAK_MARK = Dst_min × $\frac{\mathbf{STRONG\_MARK}}{\mathbf{STRONG\_STORM\_THRESHOLD}}$

With the default values, this gives: WEAK_MARK = Dst_min / 2

LOCAL_MAX_RADIUS_HOURS defines the time window used to identify a local Dst maximum during the backward search for the storm onset. A point is accepted as a local maximum if it has the highest Dst value within the surrounding ±LOCAL_MAX_RADIUS_HOURS window. The default value is **3 hours**. Increasing this parameter makes the onset detection less sensitive to short-term Dst fluctuations. However, if the value is too large, an earlier local maximum may be selected when pre-storm variations are present, resulting in an overestimation of the main phase duration.

Exclusion due to previous-storm overlap

After t_start has been estimated, the algorithm checks whether the Dst minimum of a previously detected storm occurs between t_start and t_min of the current storm. If such a previous minimum exists, the current storm is excluded from the filtered sample. This criterion prevents the algorithm from assigning a main phase interval that extends across an earlier storm minimum. In such cases, the estimated interval no longer represents the development of a single independent storm. This exclusion helps ensure that main phase duration and the subsequent analysis of relationships with solar wind parameters are calculated only for storms with a physically meaningful onset-to-minimum interval.

Disturbance filtering

After the previous-storm exclusion has been applied, the program optionally applies a disturbance filter. This filter is applied primarily to low-intensity storm events in which the Dst evolution before Dst_min is irregular, containing multiple secondary dips. Such events are still geomagnetic storms, but they are separated from the filtered storm sample because their automatic t_start determination is less reliable. In an idealized storm evolution, Dst decreases relatively continuously from the main phase onset toward Dst_min. In practice, however, many storms show irregular pre-minimum Dst variations, including secondary dips before the final minimum. For low-intensity storms, these variations can make the automatic onset detection particularly ambiguous and may produce disproportionately long main phase durations, for example several tens of hours for a moderate storm with Dst_min near −60 nT. The disturbance filter therefore separates these ambiguous events from the filtered storm sample.

DISTURBANCE_FILTER activates or deactivates this filtering step.

DISTURB_LEVEL defines the intensity range to which the disturbance filter is applied. Only storms with Dst_min shallower than DISTURB_LEVEL are tested by the disturbance filter. With the default value of **−100 nT**, the filter is applied to moderate storms.

DISTURB_DIP_COUNT defines the minimum number of secondary Dst dips required for a storm to be classified as a disturbance. If the number of dips between t_start and t_min reaches this value, the event is excluded from the filtered storm sample and added to the disturbance category. The default value is **2**.

DISTURB_DIP_RADIUS_HOURS defines the time window used to identify secondary Dst dips during disturbance filtering. Between t_start and t_min, the algorithm checks whether each Dst value is the lowest value within the surrounding ±DISTURB_DIP_RADIUS_HOURS window. If it is, the point is counted as a secondary Dst dip. The default value is **3 hours**. A larger radius makes the filter less sensitive to short-term Dst fluctuations, reducing the number of detected secondary dips and making storms less likely to be classified as disturbances.

Stage 3: Solar wind-derived parameter computation

In Stage 3, the program uses the OMNIWeb solar wind data within the main phase intervals identified after Stage 2 filtering to compute the corresponding solar wind-derived parameters for each storm. Before computing the final parameters, the program checks the OMNIWeb solar wind data for missing values during the main phase of each storm. If a single value is missing for a given parameter, it is replaced by linear interpolation using the neighboring valid values. Storms containing multiple consecutive missing values during their main phase are rejected from the final data-complete sample. This step ensures that the derived solar wind parameters are calculated only for storms with sufficiently complete input data.

For each data-complete storm, the derived solar wind parameters are computed over the main phase and include the IMF peak (IMFp), Bz peak (Bzp), solar wind speed peak (Vp), Ey peak (Eyp), and integrated Ey (Eyi). The Bz peak is defined as the most negative Bz value during the main phase, whereas the other peak quantities correspond to the maximum values reached during the same interval.